HACKATHON · STARKNET

Private OTC

Atomic peer-to-peer trades.
No escrow. No third party. No funds ever locked.

~2 MINUTES

Live demo

- Two browsers. Same `trade_id`. Opposite legs.
- One click each — **order doesn't matter**.
- Watch: balances flip atomically. Tampering reverts.

WHAT WE CHANGED IN THE PRIVACY POOL

Proof and execution, decoupled

Pool change

- Split `apply_actions` → `store_actions` + `apply_stored_actions`.
- **Proof verification** at *store* time. **Execution** at *apply* time.

Built on top (our OTC app)

- Apply **both legs atomically** in one tx.
- Use the existing `InvokeExternal` hook to assert a **conditional check** during apply → conditional trades.

WHAT YOU GET

A genuinely private OTC trade

- **Private** — amounts and tokens encrypted; only the two parties decrypt.
- **Atomic** — both legs settle in one tx, or neither does.
- **Peer-to-peer** — no third party, no escrow, no lock window.
- **Permissionless** — no identity checks. *Anyone* can submit the txs; only valid proofs settle. Bad proofs simply fail.
- **Symmetric** — both sides submit the same shape of call. No initiator role.
- **Order-independent** — second-to-join automatically settles both.

HOW — THE KEY IDEA

One proof carries both halves of the trade

- Each leg's proof commits to **(a)** transferring to the counterparty, and **(b)** checking the counterparty also transferred — scoped to the same `trade_id`.
- Naïve impl ⇒ **chicken-and-egg deadlock.**
- **Fix:** the helper contract turns the check into *"counterparty transferred me **OR** committed to transfer me here."*
- Net: a check that's **on-chain, private, and token+amount-bound.**

PROOF POINTS

It works. It can't be cheated.

ORDER-AGNOSTIC	TAMPER-EVIDENT
Alice first → settles	Wrong token → revert
Bob first → settles	Wrong amount → revert
Simultaneous → settles	Wrong recipient → revert
	Wrong `trade_id` → never pairs up

Videos show the revert + the failing assertion in the explorer.

WHAT'S NEXT

Match Maker integration

- Alice and Bob submit their conditional proofs to a **Match Maker** — *not* to each other.
- The MM adds the only missing piece: a proof of *"transfer to Alice and transfer to Bob."*
- **Alice and Bob never need to know about each other.** Discovery, matching, and settlement on the fly.
- Add an **enclave** ⇒ even the MM doesn't see the trade details.

> ***Atomic. No lock. Zero trust. Executed on the fly.***
> *Today the parties agree off-chain. Tomorrow the Match Maker does it for them — with the same privacy and atomicity guarantees.*